Neither route to a smaller cell reaches the free line

A width step lands one thousandth past it; weight tying never escapes the plateau at all.

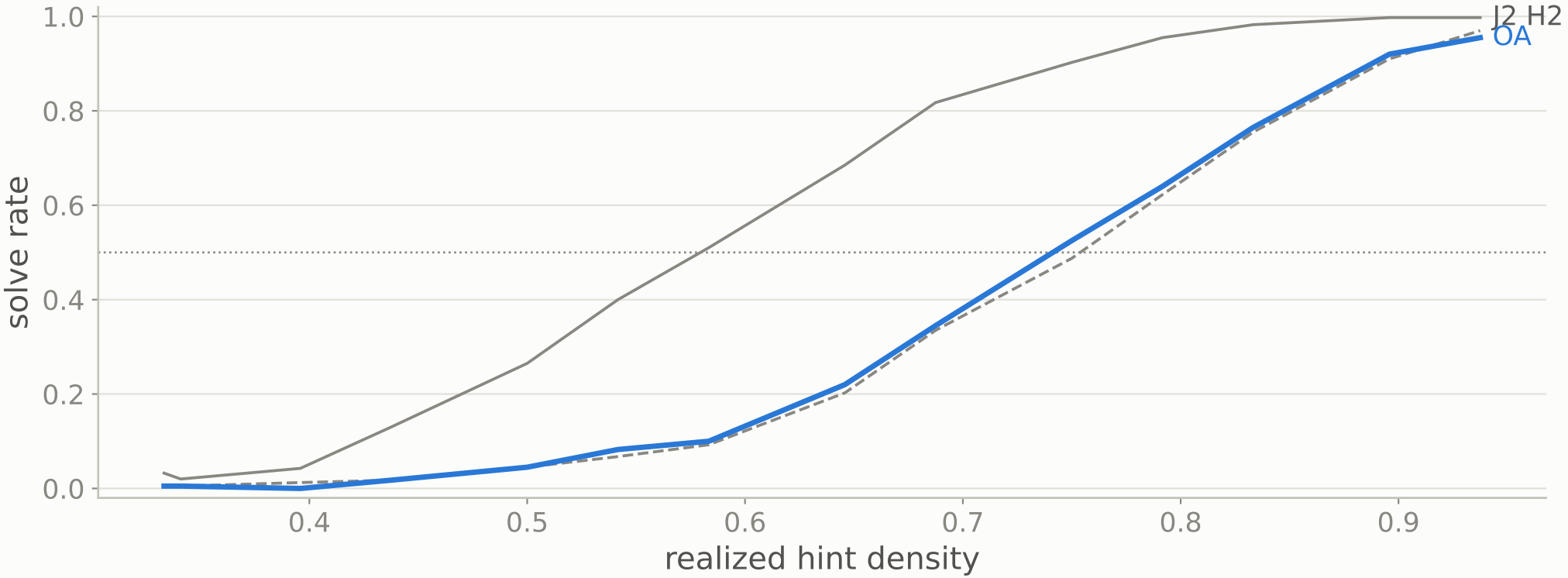
arrays 51801495 / 51815311 | scorers 51801620 / 51815318 | commit dad8929 | 3 seeds | N=48 d5 q4 --shuffle-adj --fall 240k x 50ep | 400/bin

The standing minimum is unchanged at 0.580 and 2,504,890

cell	trainable	rho_50	crossing	escapes
J2 L=16 d192	**2,504,890**	**0.580**	**3/3**	**3/3**
OA L=16 d176	2,115,930	0.741	3/3	3/3
P1 tied T=16 d192	270,010	CENSORED	1/3	0/3

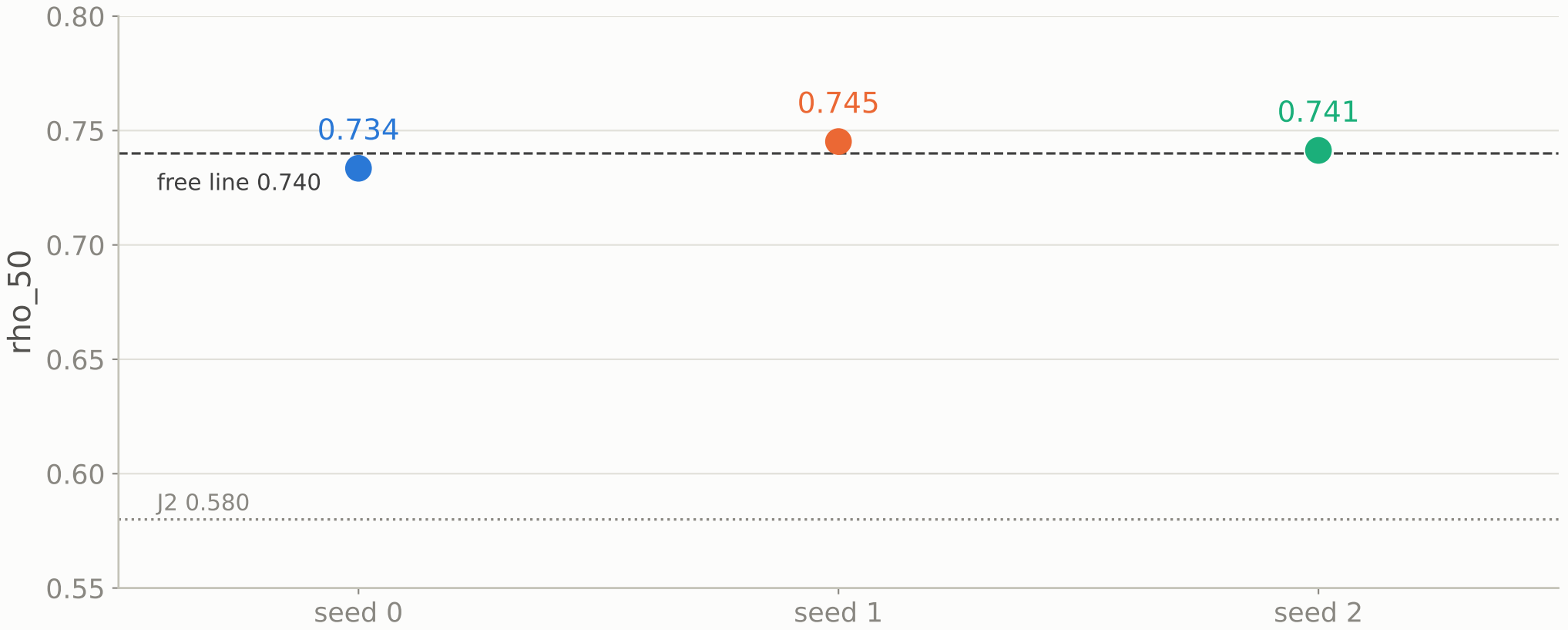
Lower rho_50 is better. Free means every seed crosses 0.5 and the median sits at or below 0.740.

OA misses the free line by 0.001, well inside its seed spread



Median 0.741 against the line at 0.740, three of three crossing. Dashed H2 and solid J2 end the measured gap.

1 of OA's three seeds is free, but the median is not



Span 0.012: 0.734 free, 0.745 past, 0.741 past. The standing rule reads the median, which lands +0.001 past the line.

Stage O's other arm was cancelled, so O0 cannot be called

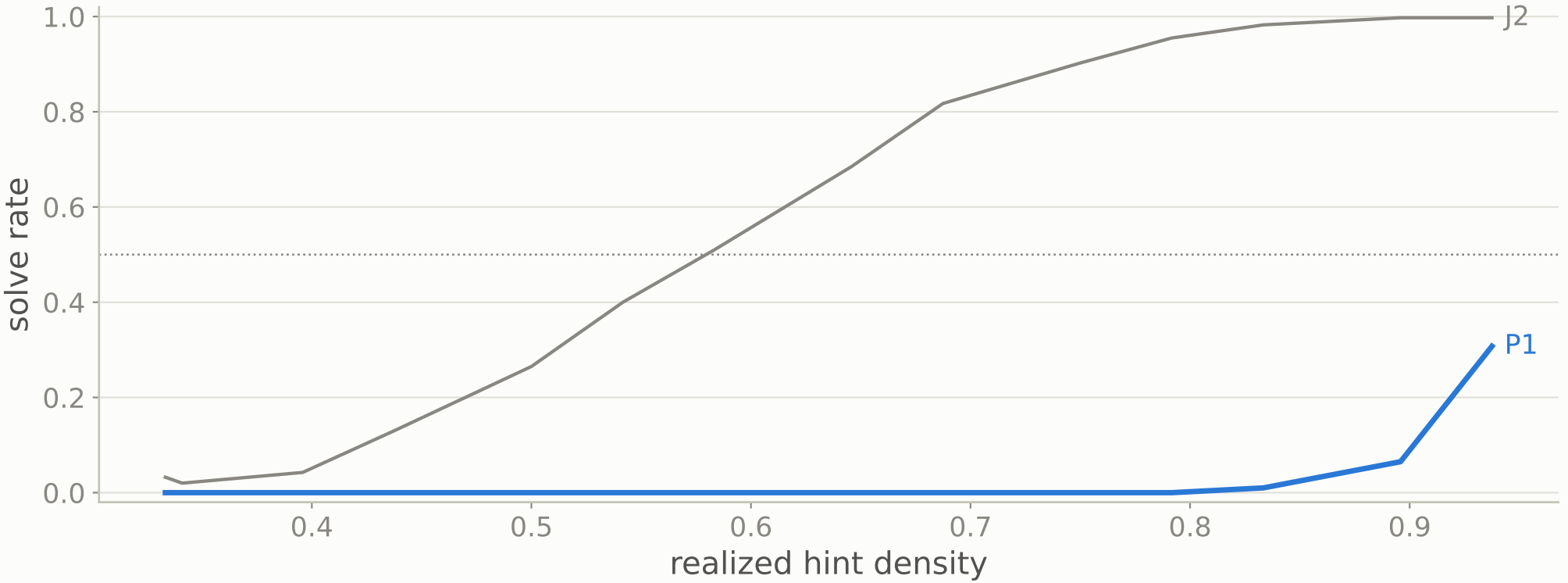
- O0 compared two SHAPES at matched size; one shape is gone
- OB (no_mlp L=20 d160) was stopped at epoch 9 of 50
- Its whole registered upside was 1.15x fewer parameters

The 1.91M-to-2.50M step now has one measured point inside it

rung	trainable	rho_50	side of the line
KA L=16 d160	1,759,738	0.757	past
H2 L=12 d192	1,908,922	0.754	past
OA L=16 d176	**2,115,930**	**0.741**	**past**
J2 L=16 d192	2,504,890	0.580	free
J1 L=12 d224	2,571,130	0.579	free

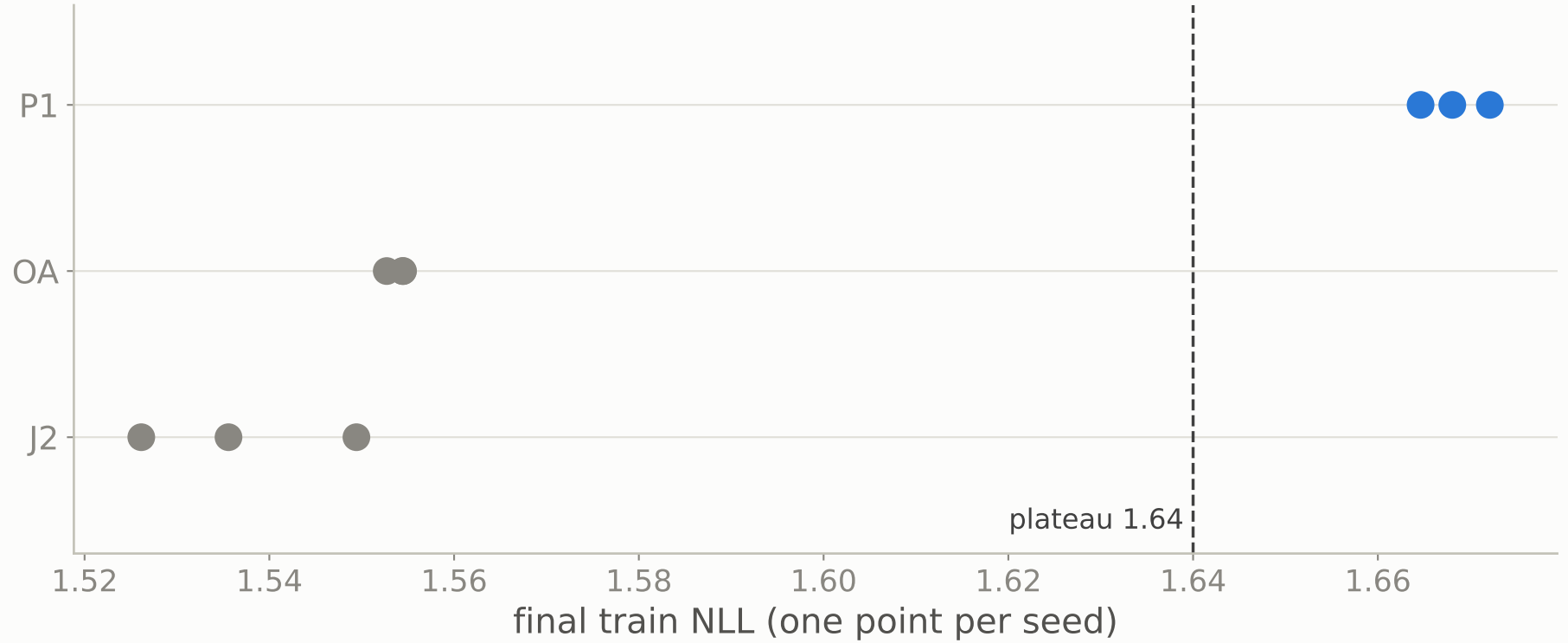
Sorted by size. Everything below 2.50M is past the line; the drop from 0.741 to 0.580 still happens in one step.

Tying at J2's shape leaves solve rate near zero until 0.8



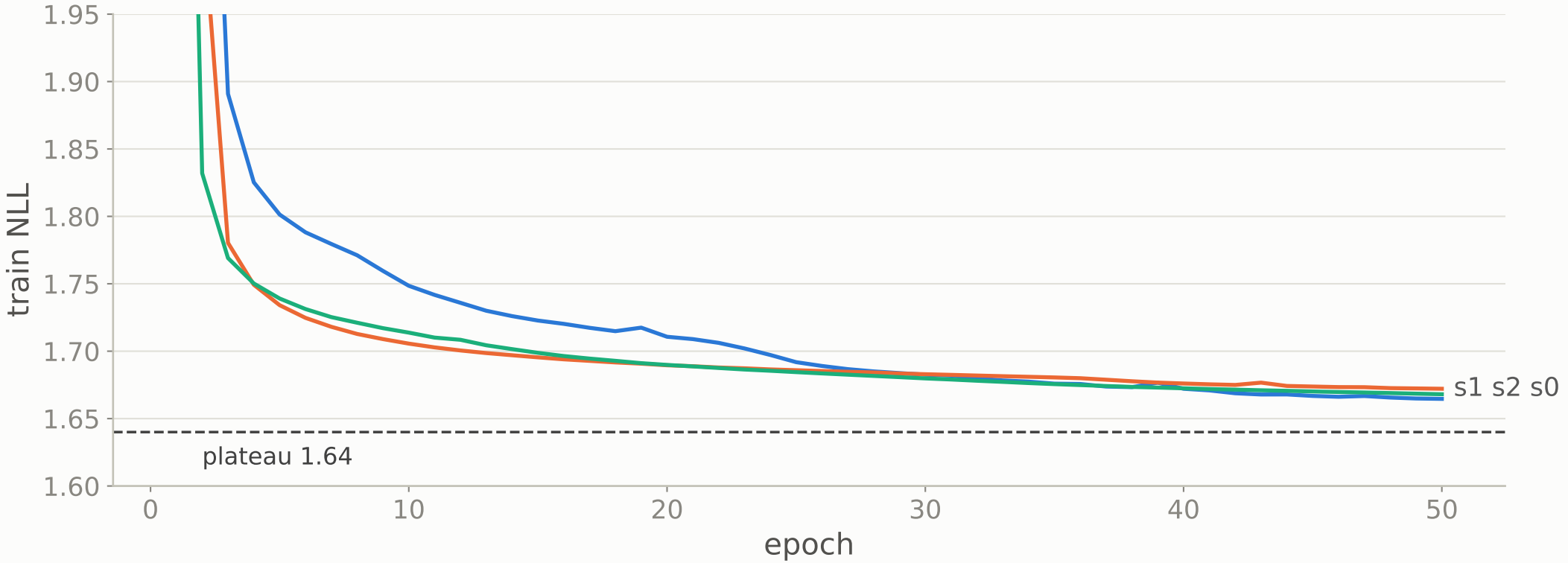
Curves pooled over seeds; one of three crosses 0.5. Same shape, budget and seeds; P1 pays the per-block term once, 270,010 against 2,504,890.

No tied seed escapes the 1.64 plateau, so P2 is falsified



Read before any level, as registered. An optimization-limited cell's rho_50 is a fact about the recipe, not about tying.

The tied curve asymptotes to 1.66 rather than destabilizing



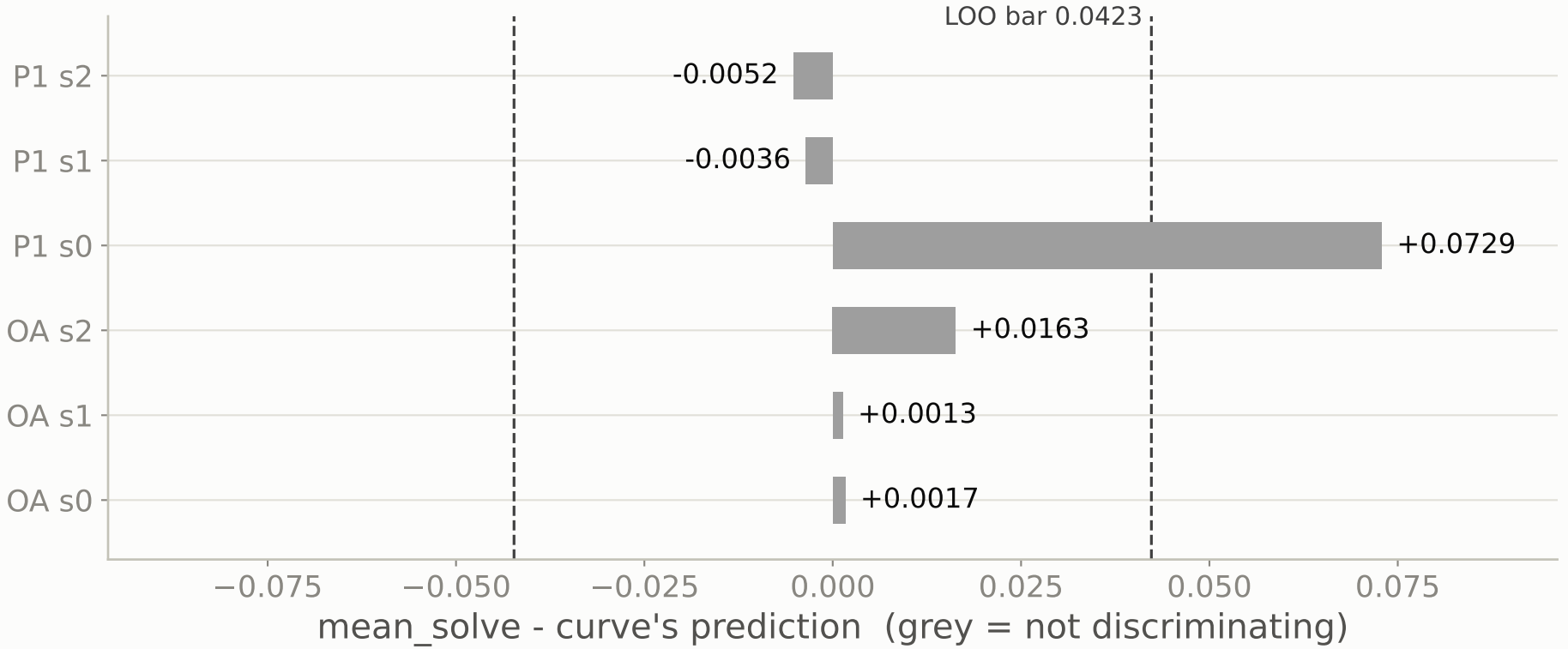
Monotone and decelerating on all three seeds, no spike at any epoch. Warmup and a lower LR treat an instability there is none of.

Accuracy orders all three cells exactly as solve rate does

cell	val token acc	mean_solve	rho_50
J2 L=16 d192	0.4940	0.5527	0.580
OA L=16 d176	0.4854	0.3320	0.741
P1 tied T=16 d192	0.4567	0.0268	CENSORED

Accuracy never censors, so it orders P1 too. Tied minus untied is -0.0372 at the same shape.

No seed's residual is discriminating: the curve cannot judge these



If tying changed the accuracy-to-solve RELATION rather than the level, this is the page where it would show.

P1's censor cannot separate tying from 270,010 being too few

- 270,010 sits below H1's 879,418, the only rung that has censored
- 'Tying costs the deduction' and '270k is too few' both predict this
- The confound was named at registration, before any number landed

Stage S straddles the censoring floor using one knob, width

cell	trainable	vs H1's 879,418	status
P1 tied T=16 d192	270,010	below	censored
S1 tied T=16 d320	613,818	below	running (51853969)
S2 tied T=16 d448	**1,088,698**	**ABOVE**	**running (51853969)**

Only d448 is above the floor, so only a censor there would be about tying. One knob changes: d_model.

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	`BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING`
	`BEGIN_ROUND SOLVED UNRESOLVED`
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	`COLOR_0 COLOR_1 COLOR_2 COLOR_3`
	the q colours (token name COLOR_n, prose 'colour')
10-57	`N0 N1 ... N47`
	one id per node, no structure in the id

GRAMMAR

`BEGIN_HINTS`	`(NODE COLOR)*`
`BEGIN_GRAPH`	`(NODE (NODE){degree})*`
`BEGIN_REASONING`	`(BEGIN_ROUND (NODE COLOR)+)*`

`SOLVED`

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 393 TOKENS

HINTS	`BEGIN_HINTS N1 COLOR_2 N3 COLOR_1 N4 COLOR_3 N6 COLOR_3 N7 COLOR_3 N8 COLOR_0 ... (59 tokens)`
GRAPH	`BEGIN_GRAPH N0 N47 N3 N26 N33 N43 N1 N23 N32 N11 N12 N38 ... (289 tokens)`
REASONING	`BEGIN_REASONING BEGIN_ROUND N5 COLOR_3 N11 COLOR_3 N14 COLOR_0 N16 COLOR_2 N17 COLOR_1 N23 ... (45 tokens)`

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 5 rounds, 19 deductions,
38 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000011, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

29 hints of 48 nodes, then 5 rounds of simultaneous deduction. Hints: N1=COLOR_2 N3=COLOR_1 N4=COLOR_3 N6=COLOR_3 N7=COLOR_3 N8=COLOR_0 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND          N5 COLOR_3 N11 COLOR_3 N14 COLOR_0 N16 COLOR_2 ...
  11 deduced
BEGIN_ROUND          N18 COLOR_0 N25 COLOR_2 N35 COLOR_1 N47 COLOR_3
  4 deduced
BEGIN_ROUND          N0 COLOR_2 N19 COLOR_0
  2 deduced
BEGIN_ROUND          N46 COLOR_2
  1 deduced
... 1 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N5, neighbours N2 N15 N9 N13 N36

their colours known before this round:

```
   [0, 1, 2]  (3 = q-1 of the 4)

   (4 known neighbours, 3 DISTINCT colours -- some repeat)

   still unknown: N2
```

only colour left: 3

```
so the trace emits  N5 COLOR_3
```

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

N=48, degree 5, 4 colours, --shuffle-adj --fall: the distribution both stages were trained and evaluated on.

T=5 rounds, the median of 40 draws from this config (range 1-19) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

Neither stage replaces the previous best on this eval

quantity	previous best J2	stage O OA	stage P P1
trainable	2,504,890	2,115,930	270,010
rho_50	**0.580**	**0.741**	**CENSORED**
seeds crossing	3/3	3/3	1/3
plateau escapes	3/3	3/3	0/3
val token acc	0.4940	0.4854	0.4567
verdict	stands	past the line	no architecture result

Same grid, same bars, same pooled s* eval: 0.580 at 2,504,890 before, 0.580 at 2,504,890 after.